

Splunk4Rookies Workshop

Lab Guide

Overview

This lab guide contains the hands-on exercises for the **Splunk4Rookies** workshop. Before proceeding with these exercises, please ensure that you have a copy of the [Splunk4Rookies slide deck](#), which will help to put into context the tasks you are carrying out.

Prerequisites

To complete these exercises, you will need your own Splunk instance. Splunk's hands-on workshops are delivered via the [Splunk Show portal](#) and you will need a splunk.com account to access this.

If you don't already have a Splunk.com account, please create one [here](#) before proceeding with the rest of the workshop.



Registering for a Splunk.com Account

To help reduce any additional identity verification checks during the registration process, please use full names and full company names where possible, avoiding nicknames and abbreviations. Please see [Splunk.com Registration Best Practices](#) for more details.

⚠ Troubleshooting Connectivity

If you experience connectivity issues with accessing either your workshop environment or the event page, please try the following troubleshooting steps. If you still experience issues, please reach out to the team running your workshop.

- **Use Google Chrome** (if you're not already)
- If the event page (i.e. <https://show.splunk.com/event/<eventID>>) didn't load when you clicked on the link, try **refreshing the page**
- **Disconnect from VPN** (if you're using one)
- **Clear your browser cache and restart your browser** (if using Google Chrome, go to: Settings > Privacy and security > Clear browsing data)
- **Try using private browsing mode** (e.g. Incognito in Google Chrome) to rule out any cache issues
- **Try using another computer** such as your personal computer - all you need is a web browser! Cloud platforms like AWS can often be blocked on corporate laptops.

Table of Contents

Exercise 1 – Access Your Lab Environment	4
Description	4
Steps	4
Enrolling in today's event if you have been given an event link:.....	4
Enrolling in today's event if you have not been given an event link:	5
Finding Your Instance Details:	7
Exercise 2 – Create an App and Add Data to Splunk	8
Description	8
Steps	8
Start Exploring Your Data	15
Description	15
Steps	15
Challenge Tasks	17
Exercise 3 – IT Operations team: Investigate successful vs unsuccessful web server requests over time	18
Description	18
Steps	18
Exercise 4 – DevOps team: Show the most common customer operating systems and which web browsers are experiencing the most failures	21
Description	21
Steps	21
Extract a New Field	21
Show the Most Common Customer Operating Systems	23
Show Which Web Browsers are Experiencing the Most Failures.....	24
Exercise 5 – Business Analytics team: Show lost revenue from the website	27
Description	27
Steps	27
Exercise 6 – Security and Fraud team: Find suspicious actors hitting both the firewall and the website	31
Description	31
Steps	31
Onboard the Firewall Data	31
Create a Table of All IP Addresses That Appear in Both the Web Server and Blocked Firewall Datasets	35
Create a Cluster Map to show the geographic location of the suspicious IP addresses	36
Challenge Tasks	37
Exercise 7 – Customize Your Dashboard.....	39
Description	39
Steps	39

Add a Custom Background Image to Your Dashboard 39

Link Your Dashboard Panels to the Global Time Picker..... 42

Challenge Task Solutions..... 44

Start Searching in Splunk 44

Exercise 6 – Security and Fraud team: Find suspicious actors hitting both the firewall and the website 45

Exercise 1 – Access Your Lab Environment

Description

You'll need a Splunk instance to do these hands-on exercises – time to get one!

In this exercise, you will obtain your own Splunk Enterprise instance using our Splunk Show portal.

Already been given your Splunk instance details?

If your workshop host has already provided you with your instance URL and login details, then **you do not need to follow the instructions in exercise 1 of this lab guide** - you can skip straight to [exercise 2!](#)

Steps

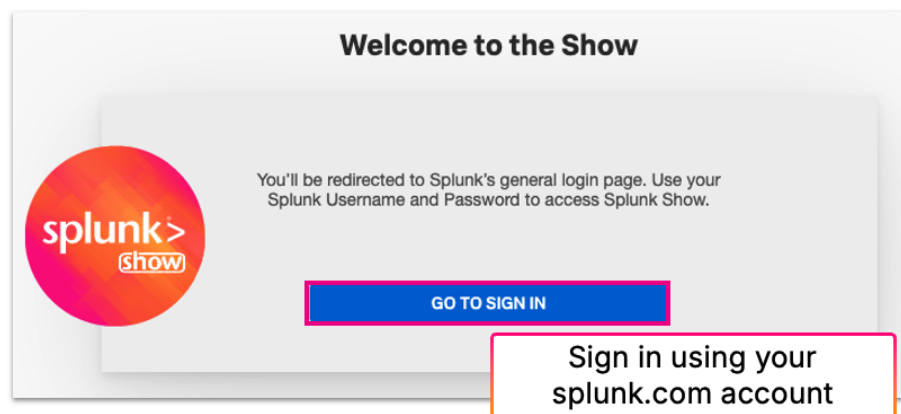
Depending on how your workshop is being run, you may have already been invited to a Splunk Show event via email or you have been given a link to a workshop event ("<https://splunk.show/...>"). If that is the case, please follow this next section [Enrolling in today's event if you've been given an event link](#).

If you're joining a public event and haven't been given a link, then please jump to [Enrolling in today's event if you haven't been given an event link](#) on the next page.

Enrolling in today's event if you have been given an event link:

1. If you **have** been given a direct link to today's workshop event ("<https://splunk.show/...>"), click on this link and it will take you to Splunk Show (<https://show.splunk.com>).

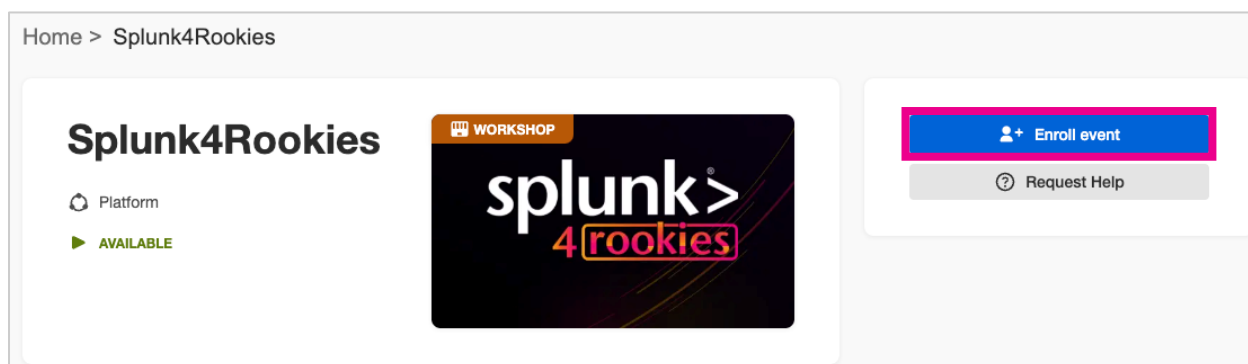
Click **GO TO SIGN IN** and log in using your **Splunk.com account**.



Don't have a Splunk.com Account?

To access our hands-on workshop events, you will need a Splunk.com account. If you don't already have a Splunk.com account, don't worry - it only takes a few minutes to create one! Please create one [here](#).

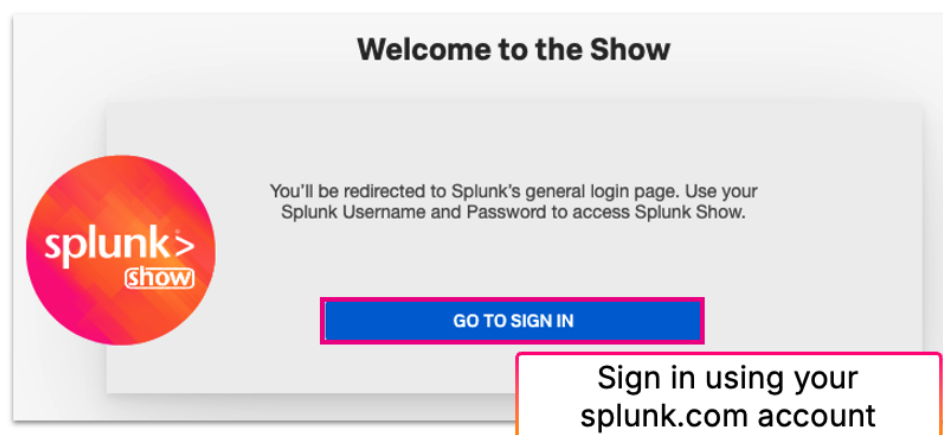
2. Once logged in you should see the event page where you can click on **Enroll event**.



(You can now jump straight to step 3 [Finding Your Instance Details](#))

Enrolling in today's event if you have not been given an event link:

1. If you **have not** been given a direct link to the event - for example, if this is a public event and you're just signing up today - please browse to <https://show.splunk.com> and click on **GO TO SIGN IN**. Log in with your **Splunk.com account**.



Don't have a Splunk.com Account?

To access our hands-on workshop events, you will need a Splunk.com account. If you don't already have a Splunk.com account, don't worry - it only takes a few minutes to create one! Please create one [here](#).

2. Once logged in to Splunk Show, scroll down to the **Events** section and find the event you have been invited to. Optionally use the 'Private' and 'Public' filters to find available events.

When you find your event, click on the **Enroll event** button to join the event.

Events All events Private Public

PUBLIC EVENT

SECURITY

Splunk SOAR Hands-On

07/25 (09:00) - 07/25 (16:57)
Europe/Prague (GMT +02:00)

Enterprise
Splunk SOAR Hands-On (Pha...

AVAILABLE Actions

Enroll event

INVITED

splunk> 4rookies

PLATFORM

Splunk4Rookies

07/25 (10:19) - 07/25 (18:09)
Europe/London (GMT +01:00)

Enterprise
Splunk4Rookies

AVAILABLE Actions

Enroll event

Find your Splunk4Rookies event

Enroll in the event

Need help enrolling?

Please ask the workshop host, who will be able to help you get enrolled!

Once you enroll the page will refresh and the event will now display 'Enrolled'.

Splunk4Rookies

Platform

AVAILABLE

Enrolled (unenroll)

This means you've enrolled successfully

Running

From 2024-07-25 at 10:19
Europe/London (GMT +01:00)

Ends in 6 hours 52 minutes

Up to 2024-07-25 at 18:09
Europe/London (GMT +01:00)

This is the date and time when your environment will be shut down

Lab environment expiration

All Splunk environments that are part of this event will automatically be shut down at the date and time specified on this screen so feel free to continue to play around with your lab environment until then!

Finding Your Instance Details:

3. Scroll down the page to the **Instances information** section and expand out the 'Splunk Enterprise' section to locate the user credentials and link to your lab environment.

The screenshot shows the 'Instances information' section with a refresh button. The 'Splunk Enterprise' section is expanded, showing a link to the instance, its status as 'RUNNING', and a table with instance details. Below this is a 'Connection Information' section with a table of credentials and a URL.

Instance ID	Termination Date	User ID
66a22627a263a1dd4776842c	6 hours 26 minutes left	-

Connection Information	
Admin Username	admin
Admin Password	
URL	https://i-09de6ec9181aefc6a.splunk.show

⚠ No connection information shown?

If you don't see any connection information displayed yet it means that your lab environment is currently starting up. Please try refreshing this view in a few minutes.

The screenshot shows the 'Instances information' section with a refresh button. The 'Splunk Enterprise' section is expanded, showing a 'Starting' status. A table with instance details is present. Below this is a 'Connection Information' section showing 'No data'. Two callout boxes provide instructions: one says 'Your instance may take up to 5 minutes to spin up so please be patient!' and the other says 'Connection information will be displayed once your environment is running'.

Instance ID	Termination Date	User ID
641871fd0b8a20001d52a5dd	23 hours 48 minutes left	-

Connection Information	
No data	

Exercise 2 – Create an App and Add Data to Splunk

Description

Splunk apps and add-ons provide customisable content and capabilities for a variety of technologies and use cases, accelerating the time it takes to get value from your data. They're also a great way to organise and share your content - such as reports and dashboards - to Splunk users. Anyone can build apps and add-ons, and today we're going to create our own app that contains a dashboard.

Since Splunk is a data platform, we'll also need to load some data in before we can do anything!

In this exercise, you will create a new app and then add some data to your Splunk Enterprise instance. We will configure Splunk to monitor some sample web server logs, which are currently being generated on the same server that Splunk is running on.

Steps

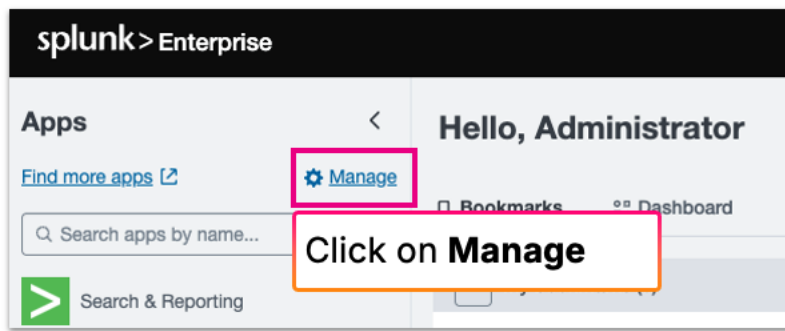
1. Browse to your Splunk instance by using the unique URL link provided in the Splunk Show event (see step 3 of [Exercise 1 - Access Your Lab Environment.](#))

Admin Username	admin	📋
Admin Password		📋
URL	https://i-07f843e7e5e12fdad.splunk.show	🔗 📋

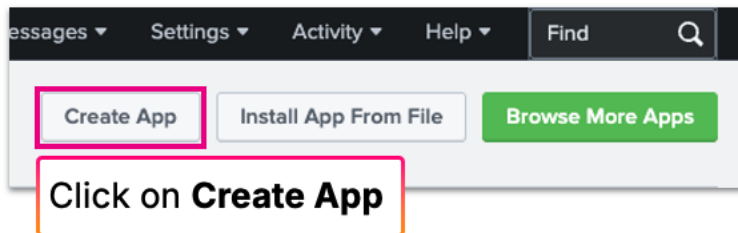
2. Log in using the credentials from the Splunk Show event.



3. On the left side of the page, under the **Apps** section, click on **Manage**.



4. On the top right corner of the screen, click on **Create app**.



5. Give your app a name and enter a folder name. Leave all other values as they are and click on **Save**.

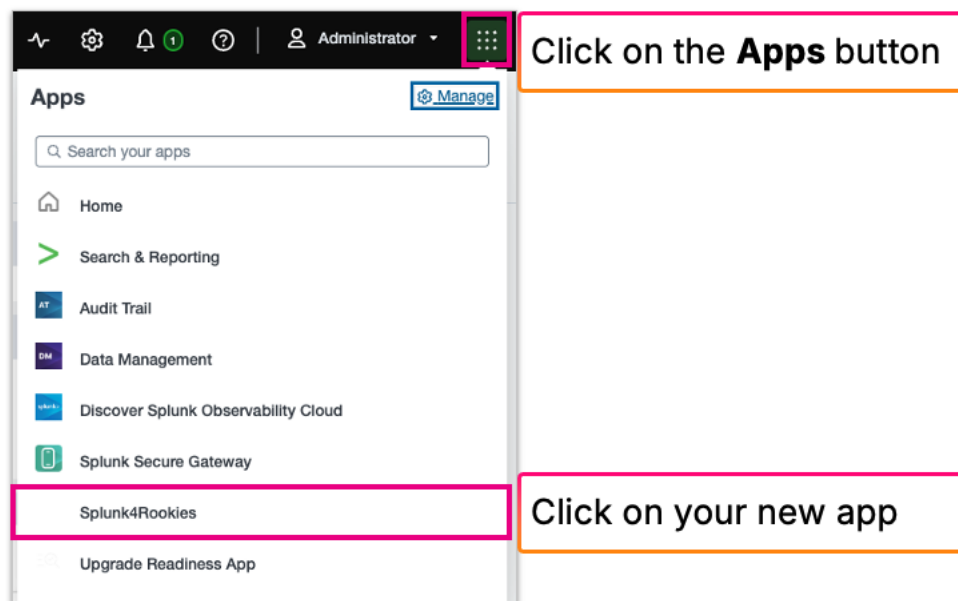
A screenshot of the 'Create App' dialog box. It has two input fields: 'Name' with the value 'Splunk4Rookies' and 'Folder name *' with the value 'splunk4rookies'. Both fields are highlighted with pink boxes. Below the 'Folder name' field, there is a note: 'This name maps to the app's directory in \$SPLUNK_HOME/etc/apps/'. At the bottom, a text box with an orange border says 'Give your app a name and specify a folder name (Note: Folder names cannot contain spaces)'. The dialog box has a close button (X) in the top right corner.

- Now that our blank app has been created, we need to select the app so that everything we do from now on will be created and saved within the new app.

To select your app, first return to the main 'Launcher' page by clicking on the Splunk Enterprise logo in the top left corner.

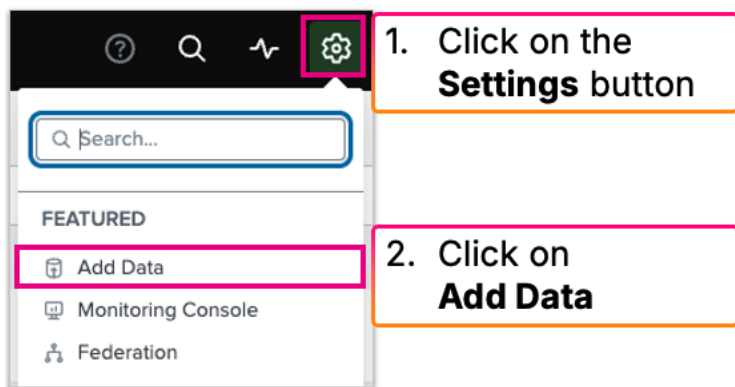
splunk>enterprise

- Click on the **Apps** menu in the top right corner of the page and select your app.

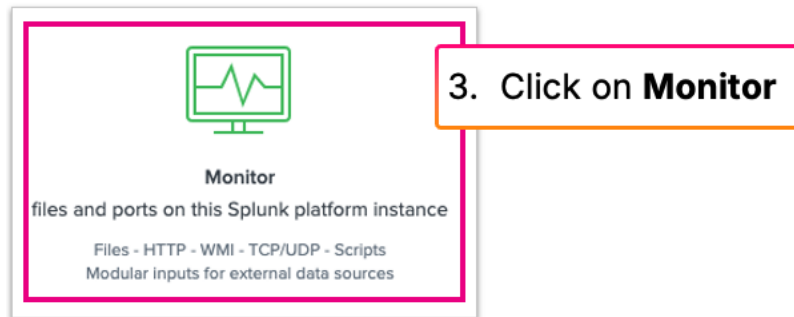


Now let's add some data!

- With our new app still selected from the dropdown list, go to **Settings > Add Data**.



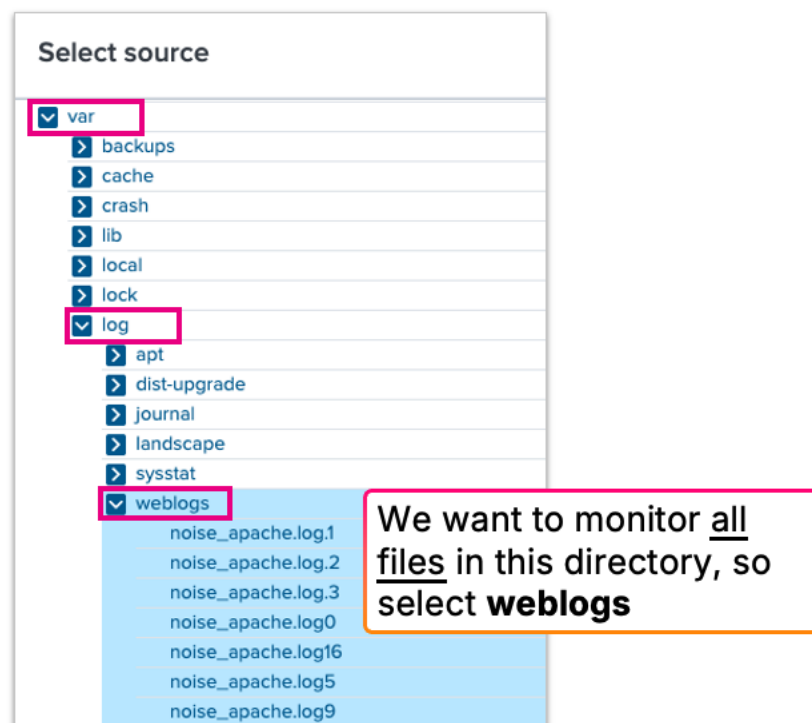
9. For this exercise we will monitor a directory, as this will allow us to pick up new data as it is generated by the web server. To do this, click on **'Monitor'**.



10. Select **'Files & Directories'** and then click **'Browse'**.



11. Browse to **/var/log** and select the **weblogs** directory. Click on **Select** to choose this directory.



12. Check that the directory path is correct (**/var/log/weblogs**) and click on **Next**.

Add Data

Select Source Input Settings Review Done

Files & Directories
Upload a file, index a local file, or monitor an entire directory.

HTTP Event Collector
Configure tokens that clients can use to send data over HTTP or HTTPS.

TCP / UDP
Configure the Splunk platform to listen on a network port.

Scripts
Get data from any API, service, or database with a script.

Splunk Assist Instance Identifier
Assigns a random identifier to each Beam node.

Systemd Journald Input for Splunk
This is the input that gets data from journald (systemd's logging).

Configure this instance to monitor files and directories for data. To monitor all objects in a directory, select the directory. The Splunk platform monitors and assigns a single source type to all objects within the directory. This might cause problems if there are different object types or data sources in the directory. To assign multiple source types to objects in the same directory, configure individual data inputs for those objects. [Learn More](#)

Data preview will be skipped, it is not supported for directories.

File or Directory ?

Includelist ?

Excludelist ?

The weblogs directory contains data that we will load into Splunk

13. Now we need to select a source type for this data. A source type determines how Splunk formats the data during the indexing process. Splunk comes with a large set of predefined source types and can often detect the source type automatically. However, for this exercise you will specify the source type.

On the **Input Settings** screen, to the right of the **Source type** section, click on **Select**.

Input Settings

Optionally set additional input parameters for this data input as follows:

Source type

The source type is one of the default fields that the Splunk platform assigns to all incoming data. It tells the Splunk platform what kind of data you've got, so that the Splunk platform can format the data intelligently during indexing. And it's a way to categorize your data, so that you can search it easily.

Click on Select

14. Click on the **Select Source Type** dropdown list and browse to **Web > access_combined**. Alternatively, you can start typing 'access' in the **filter** field and the 'access_combined' source type should appear.

Select Source Type

filter

Application

Database

Email

Log to Metrics

Metrics

Miscellaneous

Network & Security

Operating System

Structured

Uncategorized

Web

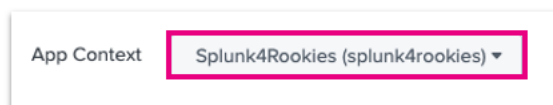
1. Click on Select Source Type

2. We're monitoring HTTP web server logs, so select Web > access_combined

access_combined
National Center for Supercomputing Applications (NCSA) combined format HTTP web server logs (can be generated by apache or other web servers)

web server

15. For the **App Context**, ensure that your new app is selected from the list.



App Context Splunk4Rookies (splunk4rookies) ▼

16. Leave all other values as default and click on **Review**.

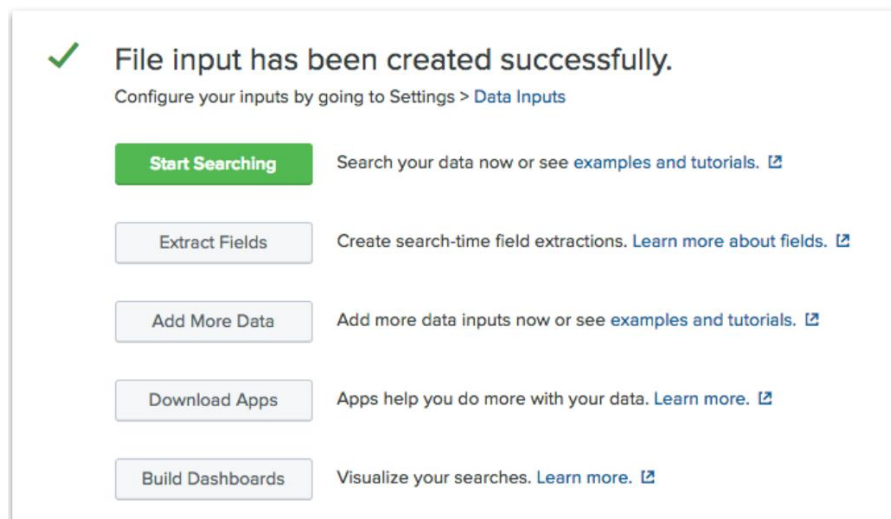
17. Review your settings and click on **Submit**.



Review

Input Type	Directory Monitor
Source Path	/var/log/weblogs
Whitelist	N/A
Blacklist	N/A
Source Type	access_combined
App Context	splunk4rookies
Host	ip-172-31-25-84
Index	default

18. You should now receive a message stating that your '**File input has been created successfully**'. Click on **Start Searching** to search the data you have just added to Splunk.



✓ **File input has been created successfully.**
Configure your inputs by going to Settings > [Data Inputs](#)

Start Searching	Search your data now or see examples and tutorials . 🔗
Extract Fields	Create search-time field extractions. Learn more about fields . 🔗
Add More Data	Add more data inputs now or see examples and tutorials . 🔗
Download Apps	Apps help you do more with your data. Learn more . 🔗
Build Dashboards	Visualize your searches. Learn more . 🔗

splunk> Enterprise

New Search

SPL Convert to SPL2

Save As ▾
Create Table View
Close

Time range: All time 🔍 🏠

✓ 153,686 events (before 08/06/2026 13:35:10.000)
No Event Sampling ▾
Job ▮ || ▮ ↶ ⬇️ ⬆️ Smart Mode ▾

Events (153,686)
Patterns
Statistics
Visualization

⌵ Timeline format ▾
↶ Zoom Out
+ Zoom to Selection
✕ Deselect
1 minute per column

▴ Format ▾
Show: 20 Per Page ▾
View: List ▾
< Prev | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | ... Next >

< Hide Fields	All Fields	i	Time	Event
SELECTED FIELDS # host 1 # source 2 # sourcetype 1		>	08/06/2026 13:35:07.000	141.146.8.66 -- [08/Jun/2026 13:35:07] "GET /cart.do?action=purchase&product_id=MCB-5&JSESSIONID=S07SL6FF3A0FF3 HTTP/1.1" 403 2068 "http://www.buttercupenterprises.com/product.screen?product_id=DfS-2" Mozilla/5.0 (Windows; WOW64) AppleWebKit/537.36 Chrome/114.0.5735.134 Safari/537.36" 809 host = show-splk10-demo-i-079b79a0e19e0cbd3 source = /var/log/weblogs/noise_apache.log26 sourcetype = access_combined
		>	08/06/2026 13:35:07.000	27.160.0.0 - [08/Jun/2026 13:35:07] "POST /cart.do?action=view&product_id=BS-2&JSESSIONID=S01SL8FF4D0FF4 HTTP/1.1" 400 2180 "http://www.buttercupenterprises.com/product.screen?product_id=PP-5" Mozilla/5.0 (iPhone; CPU iPhone OS 14_0 like Mac OS X) AppleWebKit/603.1.30 Version/14.0 Mobile/15A5370a Safari/602.1 (compatible; bingbot/2.0; http://www.bing.com/bingbot.htm)" 315 host = show-splk10-demo-i-079b79a0e19e0cbd3 source = /var/log/weblogs/noise_apache.log26 sourcetype = access_combined
		>	08/06/2026 13:35:07.000	10.2.1.44 -- [08/Jun/2026 13:35:07] "POST /product.screen?uid=19646e67-9912-4258-bc3d-c5dea3b66a99&product_id=PP-5&JSESSIONID=S04SL5FF5A0FF3 HTTP/1.1" 503 729 "http://www.buttercupenterprises.com/product.screen?product_id=CH-1" Mozilla/5.0 (Macintosh; Intel Mac OS X 10_13_6) AppleWebKit/537.36 Chrome/114.0.5735.106 Safari/537.36" 489 host = show-splk10-demo-i-079b79a0e19e0cbd3 source = /var/log/weblogs/noise_apache.log26 sourcetype = access_combined
		>	08/06/2026 13:35:07.000	141.146.8.66 -- [08/Jun/2026 13:35:07] "POST /cart.do?action=changequantity&product_id=BW-3&JSESSIONID=S03SL5FF8A0FF5 HTTP/1.1" 200 2838 "http://www.buttercupenterprises.com/product.screen?product_id=CH-1" Mozilla/5.0 (Linux; Android 13.0.0; Pixel Build/TQ3A.230705.001.A1) AppleWebKit/537.36 Chrome/106.0.5249.118 Mobile Safari/537.36" 888 host = show-splk10-demo-i-079b79a0e19e0cbd3 source = /var/log/weblogs/noise_apache.log26 sourcetype = access_combined
INTERESTING FIELDS # action 5 # bytes 100+ # clientip 67 # date_hour 5 # date_mday 1 # date_minute 60 # date_month 1 # date_second 60 # date_wday 1 # date_year 1 # date_zone 1 # file 2 # idnt 1 # index 1 # JSESSIONID 100+ # lineageat 1 # method 2		>	08/06/2026 13:35:07.000	27.96.128.0 - [08/Jun/2026 13:35:07] "GET /cart.do?action=changequantity&product_id=ZSG-2&JSESSIONID=S05SL6FF1ADFF3 HTTP/1.1" 200 903 "http://www.buttercupenterprises.com/product.screen?product_id=WPS-2" Mozilla/5.0 (Linux; Android 13.0.0; Pixel Build/TQ3A.230705.001.A1) AppleWebKit/537.36 Chrome/106.0.5249.118 Mobile Safari/537.36" 623 host = show-splk10-demo-i-079b79a0e19e0cbd3 source = /var/log/weblogs/noise_apache.log26 sourcetype = access_combined

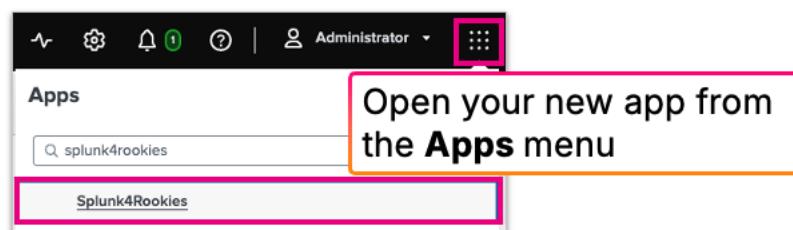
Start Exploring Your Data

Description

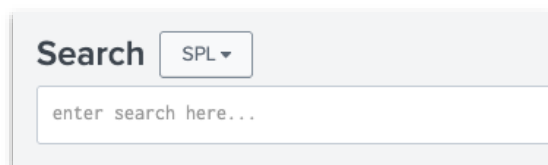
In this exercise, you will try some basic Splunk searches using the Search section of your new app.

Steps

1. Click on the **Apps** menu in the top right corner of the page and select your app.



2. To search, just type any word or phrase into the search bar and Splunk will search for all events that contain those words.



Let's try some searches!

Firstly, set the time picker (to the right of the search bar) to **Last 60 minutes**. Your environment has an event generator running in the background, which is constantly creating sample data for you to use. This data started being generated from the moment you registered for your Splunk environment, so let's stick to the last 60 minutes of data...

Try the following search:

503 purchase

This will return all events from Splunk that contain the number '503' and the word 'purchase'.

i Spaces between words in a search

In Splunk, a space between two words is an implied Boolean '**AND**', meaning that Splunk will automatically search for events containing both words – you don't need to specify it.

3. That's great, but what if there are events with the word '*purchased*', '*purchasing*', or '*purchaser*', for example? Well, we can use a wildcard asterisk (*) to search for any events containing '**503**' and any word beginning with '**pur**':

```
503 pur*
```

A wildcard is useful if we want to be a bit more flexible with what we're searching for.

4. Remember the '**AND**' operator we mentioned in step 3? Well, you can also use the other Boolean operators as well: **OR** and **NOT**. Note that these must be in UPPERCASE.

Let's try using one of these operators in a search:

```
503 (purchase OR addtocart)
```

This search will return all events containing the number '**503**' and either the word '**purchase**' or the word '**addtocart**'.

5. So far, we've just been searching for text – those numbers could appear anywhere in our data, so how do we know that we're searching the right values? Depending on our data '**503**' could be a HTTP status code, or it could be part of a session ID or a phone number.

Well, we know we're looking at web logs, so let's include field/value pairs in our search to be more specific with what we're looking for:

```
status=503 action=purchase
```

This will ensure that our results only return web server **purchase** events where the HTTP status code is '**503**'. Always specify field names where possible to ensure that your results are as accurate as possible!

Search Best Practices

In a production environment you will likely have much more data to search through than in today's workshop environment. As a best practice, always specify the index and sourcetype if you know them - it will make your searches MUCH faster!

Example:

```
index=main sourcetype=access_combined
```

For more information, please see [Write better searches](#) in the Splunk docs.

Splunk AI Assistant

You may notice an AI Assistant icon in the search bar. This is Splunk's AI Assistant — it lets you describe what you want in natural language and generates the SPL query for you. While not enabled in today's workshop environment, in production it's a powerful tool for building queries quickly. Today you're learning the fundamentals of SPL that will help you understand and validate what the AI Assistant generates.

Challenge Tasks

Q1. How can we find events with a status of 200 that are not purchase events?

Q2. How can we find events where someone had an error when trying to either add an item or remove an item from their cart? (**Hint:** A HTTP status code of 200 means the transaction was successful. A code of 400 or higher usually means that a failure occurred.)

Challenge Task Solutions

The challenge task solutions are at the [end of this document](#).

Exercise 3 – IT Operations team: Investigate successful vs unsuccessful web server requests over time

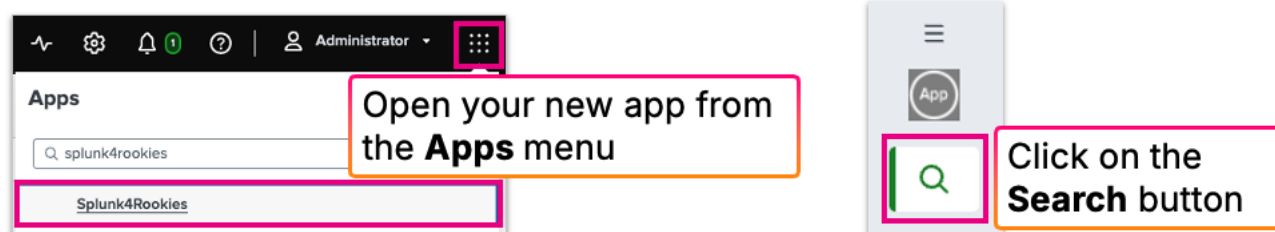
Description

The IT Operations team currently has no visibility of failures on the Buttercup Enterprises website.

In this exercise, you will produce a dashboard panel for the IT Operations team, showing website successes vs failures over time.

Steps

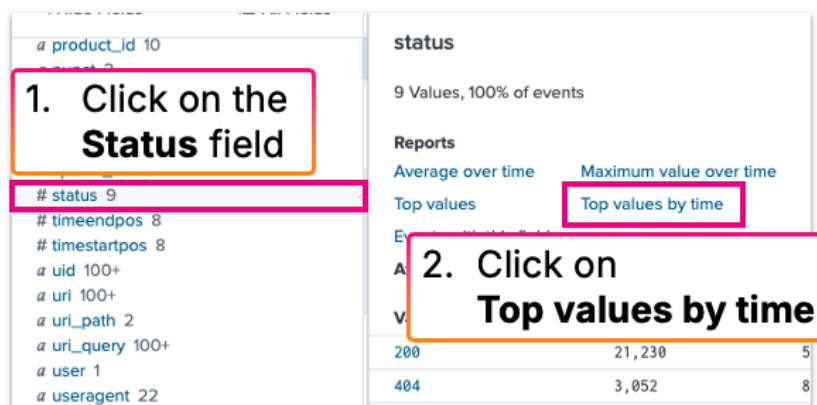
1. To start a new search, first make sure your app is selected from the Apps menu list and then click on the Search icon in the left-hand menu.



2. Search the **main** index (i.e. the default index) for all web server events over the **Last 60 minutes**:

```
index=main sourcetype=access_combined
```

3. Scroll down the page and find the **status** field. Click on the field name to display the field window and select **Top values by time**.

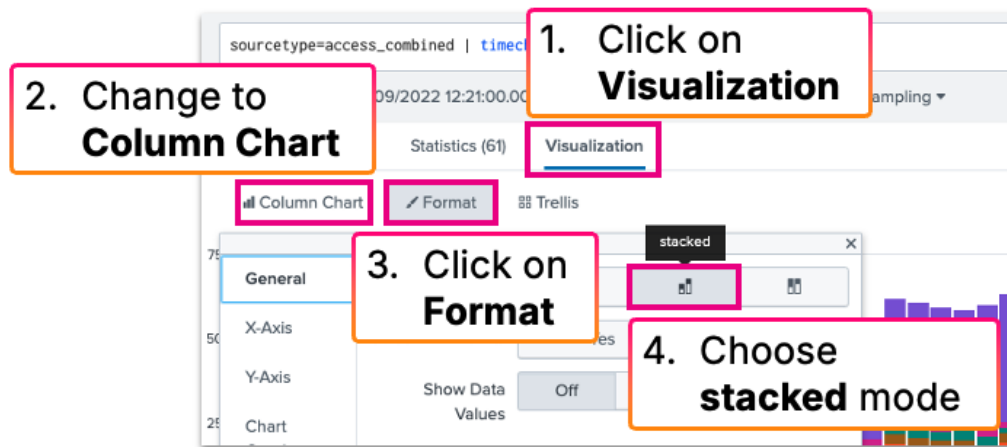


Splunk will automatically populate your search as follows:

```
index=main sourcetype=access_combined | timechart count by status limit=10
```

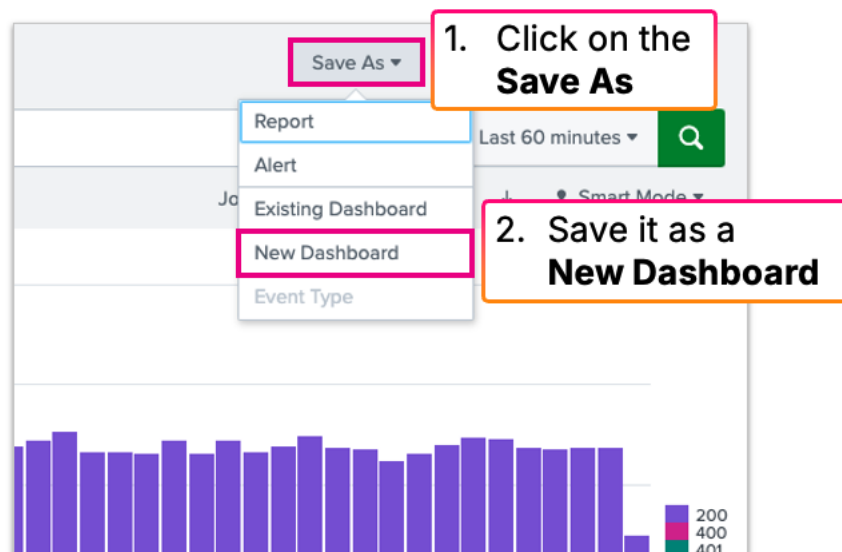
4. A chart will display on the **Visualization** tab. Change the visualization to a **Column Chart**.

Click on **Format** and then on the **General** tab to change the **Stack Mode** to '**stacked**'. Feel free to play around with the formatting until you're happy with the visualization.



5. Now that we have a nice chart visualization, let's add it to a new dashboard so we can share this information with the business.

In the top right corner of the screen, go to **Save As > New Dashboard**.



6. On the **Save Panel to New Dashboard** screen, give your dashboard a suitable title and optionally a description too. If you can't think of a name for your dashboard, call it '**Buttercup Enterprises**, or something else meaningful to you.

Choose your dashboard type. For today's workshop we will use **Dashboard Studio**.

For your layout mode, select **Absolute**.

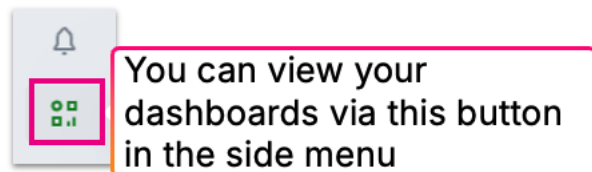
Give your panel a title – something that describes what this chart is showing, such as '**IT Ops - Web Server Status Codes Over Time**'.

The screenshot shows the 'Save Panel to New Dashboard' dialog box with the following fields and options:

- Dashboard Title:** Buttercup Enterprises (with a callout: 1. Give your dashboard a title and description)
- Description:** Dashboard for the Buttercup Enterprises team
- Permissions:** Shared in App
- Dashboard type:** Dashboard Studio (with a callout: 2. Select Dashboard Studio)
- Select layout mode:** Absolute (with a callout: 3. Select Absolute)
- Panel Title:** IT Ops - Web Server Status Codes over Time (with a callout: 4. Give your panel a name – make it clear what the chart is showing)
- Visualization Type:** Bar Chart
- Buttons:** Cancel, Save to Dashboard

7. Click on **Save to Dashboard** and then **View Dashboard**.

Congratulations - you've just created a Splunk dashboard with your first panel! Anytime you want to access a dashboard, click on the **Dashboards** icon in the side menu bar and select the dashboard you wish to view. Go ahead – give it a try!



Exercise 4 – DevOps team: Show the most common customer operating systems and which web browsers are experiencing the most failures

Description

In this exercise, you will need to extract a new field from your events in order to create the report we need. To accomplish this, we will use Splunk's field extractor wizard.

Custom field extractions are useful in a variety of scenarios, such as:

- When you have custom data and Splunk did not recognise/extract a particular field that you need
- When you need to extract a particular part of an event in order to be able to search/report on that value

Steps

Extract a New Field

1. If you don't already see the search page displayed click on the Search icon in the side menu.

Search for all web server events over the **Last 60 minutes**:

```
index=main sourcetype=access_combined
```

2. Expand out one of the events by clicking on the arrow (>) to the left of the event timestamp. Click on the **Event Actions** dropdown list and select **Extract Fields**:

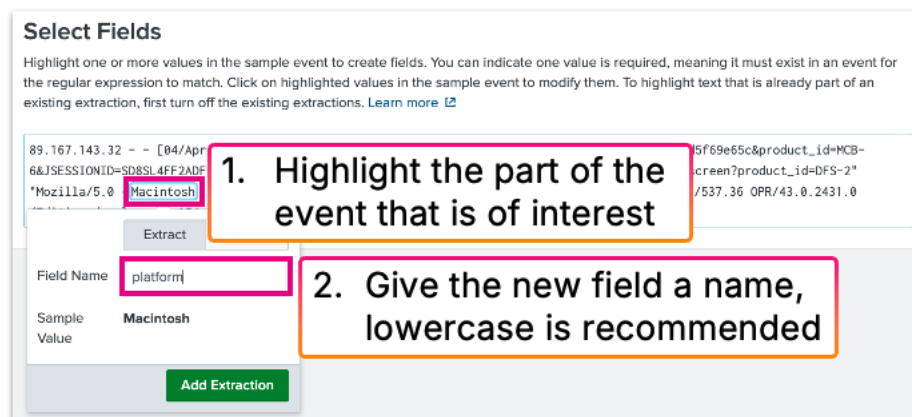


3. We have two options for extracting fields: Regular Expression or Delimiters. For this exercise, we will choose Regular Expression. Click on **Regular Expression** and then click on **Next**.

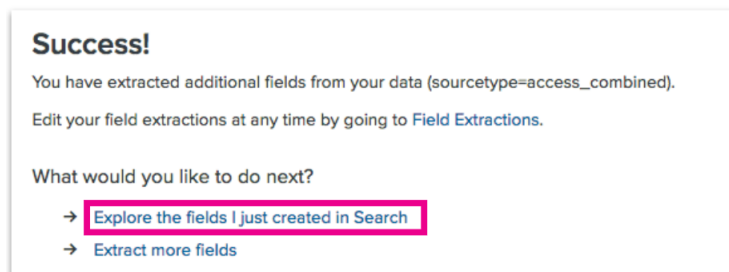


4. You will now be presented with a sample event from which to extract your field. For this exercise, we will need to extract the platform (operating system) information from each event so we can report on it. Look for the platform/operating system information in your event (e.g. Linux, Macintosh, Windows, etc.) contained in the useragent string towards the end of the event and highlight it.

Give the new field the following name: **platform** (field names are case sensitive, so be sure to use all lowercase letters for this to make your life easier!)



5. Click on **Add Extraction** and then click on **Next**.
6. Click on **Next** again to reach the **Save** screen. On the Save screen, click on **Finish** to save your new field extraction.
7. You should now see a **Success!** page. Click on **Explore the fields I just created in Search**.



8. Splunk will show you search results for all of your web server data over the last 24 hours. Scroll down the page and look for your new field listed on the left – you can now use it in your searches!

Look for your new field in the list!

Values	Count	%
Macintosh	11,597	31.955%
Windows	11,478	31.627%
Linux	6,619	18.238%
iPhone	6,598	18.18%

Show the Most Common Customer Operating Systems

Now that we have our new field, we can use it to report for the DevOps team!

9. Search for all web server events over the **Last 60 minutes**:

```
index=main sourcetype=access_combined
```

10. Scroll down the page and find the **platform** field that you just extracted. Click on the field name to display the field window, and then select **Top values**.

1. Click on the **platform** field

2. Click on **Top Values**

Splunk will automatically populate your search as follows:

```
index=main sourcetype=access_combined | top limit=20 platform
```

11. Select the **Visualization** tab if not already displayed and change the visualization to a **Bar Chart**.

1. Click on **Visualization**

2. Change the visualization to a **Bar Chart**

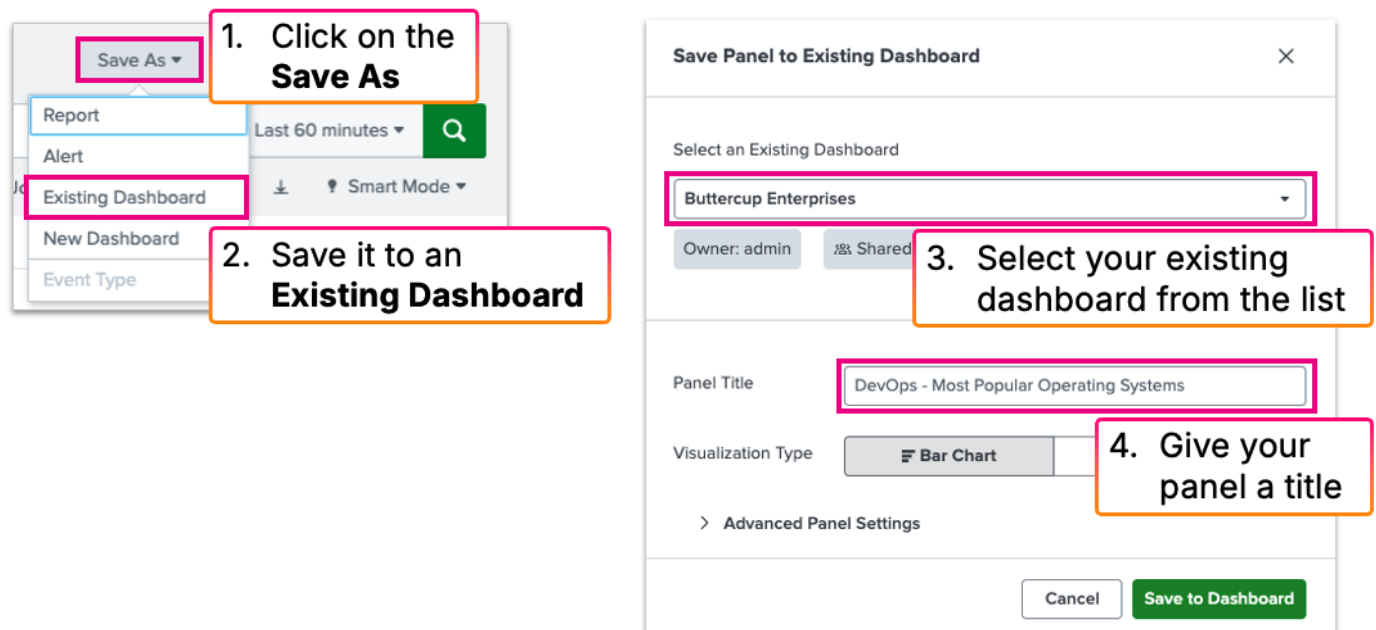
i Tip for cleaning up your chart

You can optionally add `showperc=f` to the `top` command to remove the 'percent' column from the table of statistics:

```
index=main sourcetype=access_combined | top limit=20 platform showperc=f
```

This will help to keep the chart nice and clean when we view it on our dashboard later.

- When you're happy with your chart, save it to an **'Existing Dashboard'** and select the dashboard you previously created from the list. Finally, give the dashboard panel a suitable title, such as **'DevOps - Most Popular Operating Systems'** and click on **Save to Dashboard**.



Show Which Web Browsers are Experiencing the Most Failures

One DevOps use case down, one more to go! We now need to report on failures by web browser.

- Search for all web server events over the **Last 60 minutes**:

```
index=main sourcetype=access_combined
```

- Add a search filter to return only events with a status code of 400 or higher (an event with a status value of 400 or higher is considered a failure of some kind.)

```
index=main sourcetype=access_combined status>=400
```


15. Scroll down the page and find the **useragent** field (Note: 'useragent' is a field containing information about the web browsers that are interacting with our website.) Click on the field name to display the field window and then select **Top values by time**.



Splunk will automatically populate your search as follows:

```
index=main sourcetype=access_combined status>=400
| timechart count by useragent limit=10
```

16. Select the **Visualization** tab if not already displayed and change the visualization to an **Area Chart**.

To make your chart cleaner, limit your output to the top 5 useragents by changing the "limit" to 5 in your search.

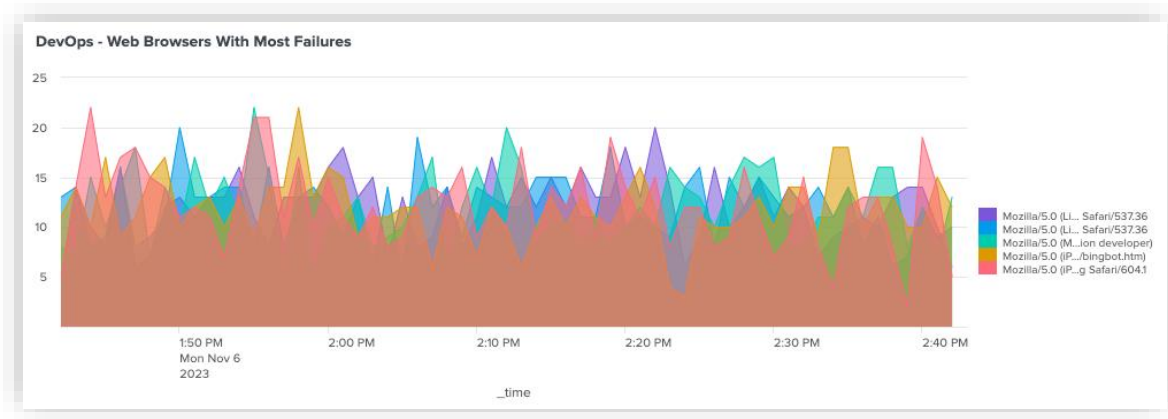
```
index=main sourcetype=access_combined status>=400
| timechart count by useragent limit=5
```

i Tip for cleaning up your chart

You can optionally add `useother=f` to the `timechart` command to remove the 'OTHER' value from your chart:

```
index=main sourcetype=access_combined status>=400
| timechart count by useragent limit=5 useother=f
```

When you're happy with your chart, add it to your dashboard and give the panel a title such as '**DevOps - Web Browsers with Most Failures**'.



Note: Remember to add it to your existing dashboard rather than creating a new one!

Exercise 5 – Business Analytics team: Show lost revenue from the website

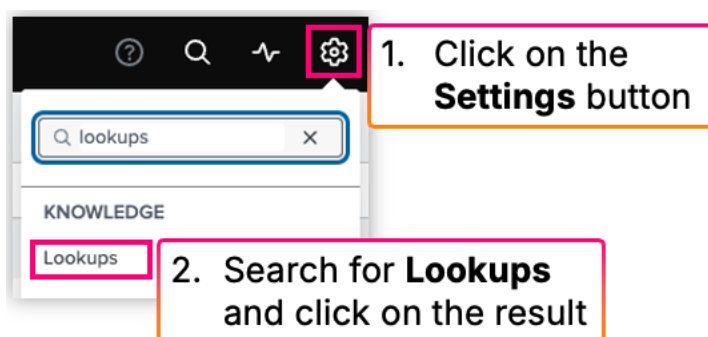
Description

Buttercup Enterprises does not have a way of seeing lost revenue from the website in real-time and the senior managers would like to track lost revenue trends throughout the day via a dashboard.

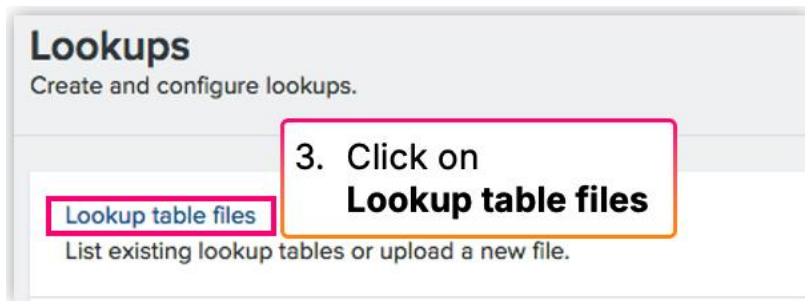
In this exercise, we will create a Single Value visualization that shows lost revenue from the company website and add this to our dashboard.

Steps

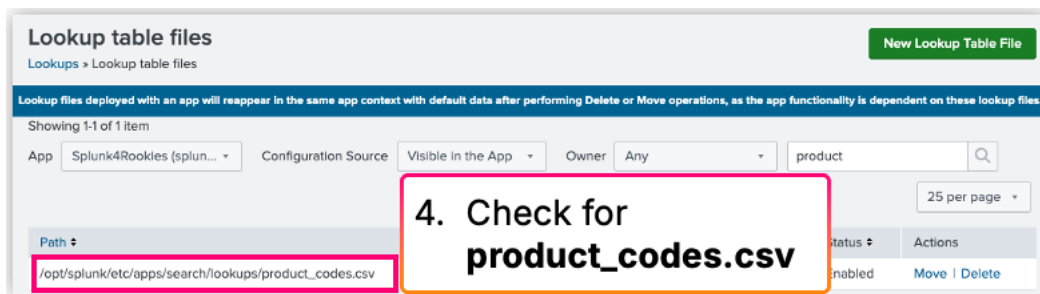
1. Go to **Settings > Lookups**.



2. Click on **Lookup table files**.



Check that the '**product_codes.csv**' file exists in your environment.



- Return to your app and make sure you are on the **Search** view.



You may want to view the contents of the lookup file to familiarise yourself with the fields and values that it contains. To do this, use the `inputlookup` command along with the name of your lookup file:

```
| inputlookup product_codes.csv
```

The resulting table should look like this:

category ↕	product_description ↕	product_id ↕	product_name ↕	product_price ↕
Clothing	Plush mascot slippers with non-slip soles. Now with 100% fewer regex requirements to put on.	BS-2	Buttercup Slippers	25.7
Books	Our caped hero indexes the city's data into perfectly time-sliced buckets - villains never escape his retention policy.	MCB-5	Mad Comics- Bucket Boy	12.7
Books	The Buttercup mascot's origin story. Spoiler: it involves a tragic accident with a misconfigured field extractions.	MCB-6	Mad Comics- Buttercup	12.7
Books	Lightning-fast and lightly armoured, our hero ships every event straight to the indexer. No data left behind.	MCF-3	Mad Comics- Forwarder	12.7
Books	A flavourful approach to data ingestion. Pairs surprisingly well with strong coffee and an unfinished correlation search.	ZSG-2	Zesty Splunker Guide	15.21
Clothing	Officially licensed Buttercup mascot costume. Warning: wearer becomes the most-searched object in any room they enter.	CM-1	Costume- Mascot	97.5
Snacks	Crispy on the outside, suspiciously fast on the inside. Not actually food. Please do not eat.	DFS-2	Deep-Fried Searches	22.75
Gifts	Soft, huggable plush in the shape of the Splunk pivot icon. For when you need emotional support during a complex search.	PP-5	Pivot Plushie	9.99
Clothing	Silicone wristband. Doubles as a discreet reminder that you forgot to widen the time picker.	BW-3	Buttercup Wristband	9.99
Gifts	Stress ball shaped like a tiny Splunk indexer. Squeeze when search runtimes exceed expectations.	WPSS-2	Workshop Participation Squishy	4.99

- Now that you've checked the lookup file, you can use the `lookup` command to extract the **product_price** field from the csv file and add it to the web server purchase events by running the following search over the **Last 60 minutes**:

```
index=main sourcetype=access_combined action=purchase
| lookup product_codes.csv product_id
```

You will notice that a **product_price** field now appears under the extracted fields on the left side of the page, along with a couple of other new fields: 'category' and 'product_name'.

The screenshot shows the Splunk interface with a list of extracted fields on the left and a detailed view of the 'product_price' field on the right. A callout box highlights the 'lookup' command in the search bar, stating: 'The **lookup** command retrieves the **product_price** field from our csv file'. The 'product_price' field is selected, and the 'Reports' section shows various options like 'Average over time', 'Maximum value over time', 'Minimum value over time', 'Top values', 'Top values by time', and 'Rare values'. The 'Events with this field' section shows statistics: Avg: 22.401588823904426, Min: 4.99, Max: 97.5, Std Dev: 25.675604949880277. A table below shows the distribution of values.

Values	Count	%
12.7	10,892	30.191%
9.99	7,268	20.146%
22.75	3,676	10.189%
4.99	3,606	9.995%

Splunk is pulling this new data from the csv file we specified using the **product_id** field, which exists in our data. You can now use these additional fields in your searches!

5. We now need to customise our search to focus on **failed purchase events**, since this is what we need to measure to calculate lost revenue. To do this, add a search filter to find events where the status is **400** or greater (i.e. an error of some kind has occurred.)

```
index=main sourcetype=access_combined action=purchase status>=400  
| lookup product_codes.csv product_id
```

6. Finally, we need to calculate the total of the **product_price** field for all of these failed purchase events **over time**. To do this we will use the **timechart** command along with the **sum** function.

The **sum** function returns the sum of the values of a field, so we need to tell it the field we want it to work with. We will specify **product_price** as knowing the sum of this field will tell us how much total revenue we have lost.

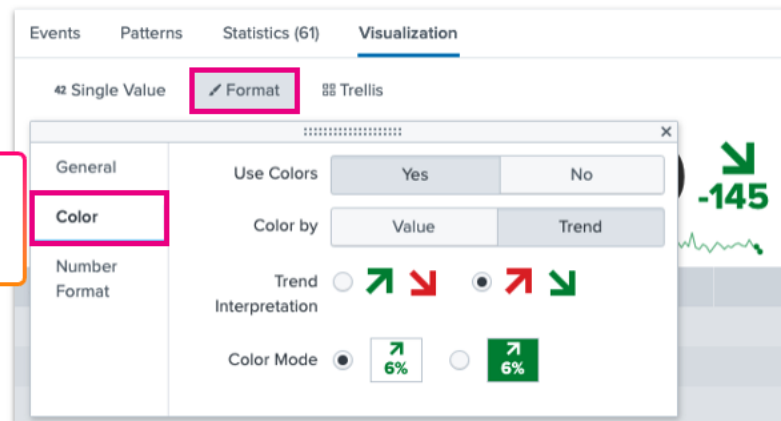
```
index=main sourcetype=access_combined action=purchase status>=400  
| lookup product_codes.csv product_id  
| timechart sum(product_price)
```

i Want to learn more about SPL?

Check out Splunk's [Search Reference](#) documentation page for a list of all the search commands and functions, along with complete syntax, descriptions, and examples of how to use them!

7. Select the **Visualization** tab if not already displayed and change the visualization to a **Single Value** visualization.

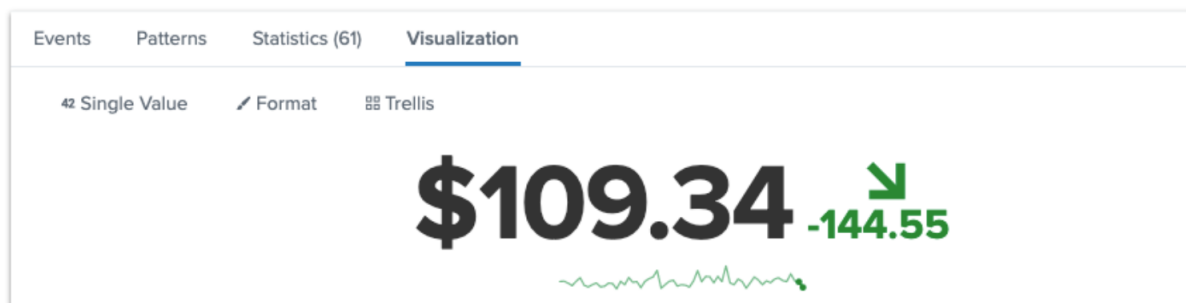
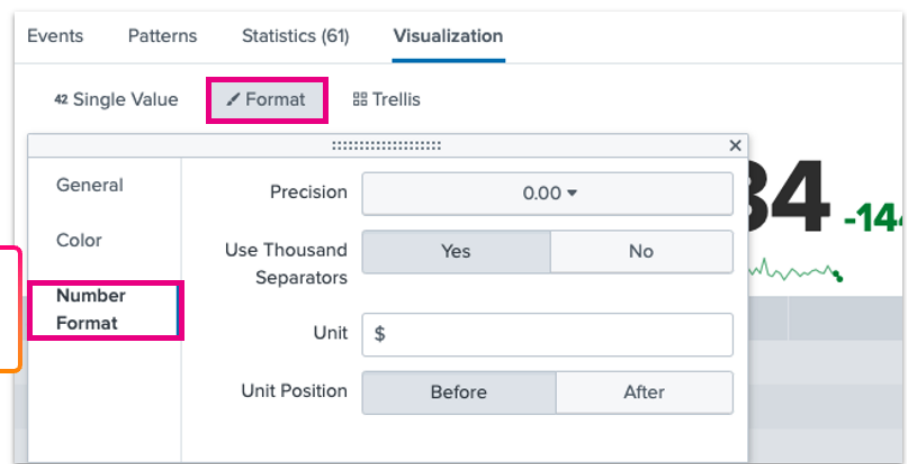
Click on the **Color** tab and try some of the formatting options!



Click on **Format** and use the side tabs to change the formatting options. Try adding some color!

8. Click on **Number Format** and add a currency unit symbol (£, \$ or €) to make it clear that it's a monetary value.

Click on the **Number Format** tab and try some of the formatting options!



Once you're happy with the visualization, add it to your dashboard and give the panel a title such as '**Business Analytics - Lost Revenue**'.

Exercise 6 – Security and Fraud team: Find suspicious actors hitting both the firewall and the website

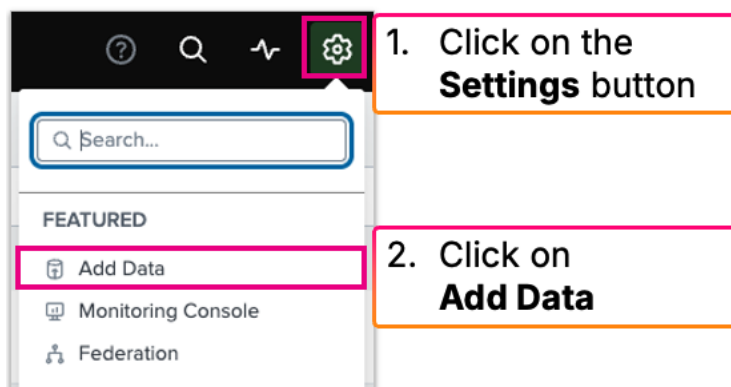
Description

Buttercup's perimeter firewall – a **Cisco ASA** – blocks suspicious IPs every day. The Security and Fraud team want to find IPs that the firewall blocked and hit the website – likely malicious traffic. We will onboard the firewall log, find the IPs that appear in both sources, and plot them on a map.

Steps

Onboard the Firewall Data

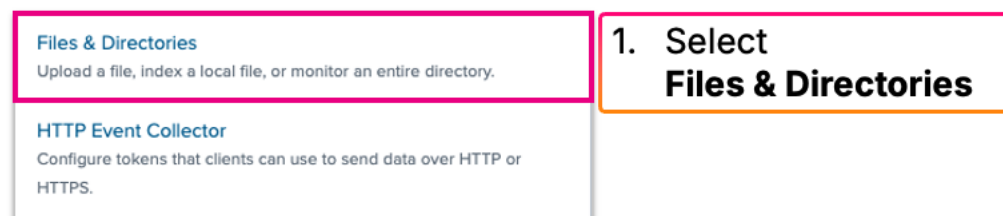
1. With our new app still selected from the dropdown list, go to **Settings > Add Data**.



2. Click on '**Monitor**'.



3. Select '**Files & Directories**' and then click '**Browse**'.



File or Directory ? Browse

On Windows: c:\apache\apache.
\\hostname\apache\apache.error

2. Click on **Browse**

- Browse to **/var/log** and select the **firewall** directory. Click on **Select** to choose this directory.

Select source

- ☒ var
 - backups
 - cache
 - crash
 - lib
 - local
 - lock
 - ☒ log
 - apt
 - dist-upgrade
 - ☒ firewall
 - cisco_asa.log11
 - cisco_asa.log9

- Check that the directory path is correct (**/var/log/firewall**) and click on **Next**.
- On the **Input Settings** screen, to the right of the **Source type** section, click on **Select**.

Input Settings

Optionally set additional input parameters for this data input as follows:

Source type

The source type is one of the default fields that the Splunk platform assigns to all incoming data. It tells the Splunk platform what kind of data you've got, so that the Splunk platform can format the data intelligently during indexing. And it's a way to categorize your data, so that you can search it easily.

Automatic **Select** New

Click on **Select**

- Click on the **Select Source Type** dropdown list and search for 'cisco:asa'.

Automatic Select New

Select Source Type ▼

asa

- cisco:asa
Output produced by the Cisco Adaptive Security Appliance (ASA) Firewall
- cisco_asa

Select **cisco:asa**

- For the **App Context**, ensure that your new app is selected from the list.

App Context Splunk4Rookies (splunk4rookies) ▼

9. For the **Index** click on **Create a new index** and call it **firewall**.

Index: Default ▾ **Create a new index**

**Click on
Create a new index**

New Index

General Settings

Index Name: **firewall** **Name it firewall**
Set index name (e.g., INDEX_NAME)

Index Data Type: **Events** Metrics
The type of data to store (event-based or metrics).

Home Path: optional
Hot/warm db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/db).

Cold Path: optional
Cold db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/colddb).

Thawed Path: optional
Thawed/resurrected db path. Leave blank for default (\$SPLUNK_DB/INDEX_NAME/thaweddb).

Data Integrity Check: **Enable** Disable
Enable this if you want Splunk to compute hashes on every slice of your data for the purpose of data integrity.

Max Size of Entire Index: 500 GB ▾
Maximum target size of entire index.

Save **Cancel**

Click on **Save** to create the new index.

10. Leave all other values on the **Input Settings** page as default and click on **Review**.

11. Review your settings and click on **Submit**.

Review

Input Type Directory Monitor
Source Path /var/log/firewall
Includelist N/A
Excludelist N/A
Source Type cisco:asa
App Context splunk4rookies
Host show-splk10-demo-i-01dfea53c5bc7433d
Index firewall

12. You should now receive a message stating that your **'File input has been created successfully'**.

Click on **Start Searching** to search the data you have just added to Splunk.

 **File input has been created successfully.**
Configure your inputs by going to Settings > [Data Inputs](#)

Start Searching

Search your data now or see [examples and tutorials](#). 

Extract Fields

Create search-time field extractions. [Learn more about fields](#). 

Add More Data

Add more data inputs now or see [examples and tutorials](#). 

Download Apps

Apps help you do more with your data. [Learn more](#). 

Build Dashboards

Visualize your searches. [Learn more](#). 

You should now see the new cisco:asa firewall data being shown in Splunk.

splunk>Enterprise

     | Administrator 

New Search SPL Convert to SPL2 Save As Create Table View Close

index=firewall Time range: Last 60 minutes  

✓ 18,149 events (19/06/2026 15:09:00.000 to 19/06/2026 16:09:27.000) No Event Sampling

Job      Smart Mode

Events (18,149) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

1 minute per column

Format Show: 20 Per Page View: List

Prev 1 2 3 4 5 6 7 8 ... Next

Hide Fields

All Fields

SELECTED FIELDS

a action 2

a host 1

a source 2

a sourcetype 1

INTERESTING FIELDS

a access_group 1

a acl 1

a Cisco_ASA_action 2

Cisco_ASA_message_id 2

a Cisco_ASA_vendor_action 2

a clientip 94

a communication_protocol 1

date_hour 2

date_mday 1

date_minute 60

date_month 1

date_second 60

date_wday 1

date_year 1

i	Time	Event
>	19/06/2026 16:09:24.000	Jun 19 16:09:24 fw-01 : %ASA-4-106023: Deny tcp src outside:74.125.19.106/7216 dst inside:10.20.30.41/80 by access-group "outside_access_in" [0x0, 0x0] action = blocked host = show-splk10-demo-i-01dfea53c5bc7433d source = /var/log/firewall/cisco_asa.log6 sourcetype = cisco:asa
>	19/06/2026 16:09:24.000	Jun 19 16:09:24 fw-01 : %ASA-6-302013: Built inbound TCP connection 1082913053 for outside:175.45.178.19/33044 (165.227.85.118/19231) to inside:192.168.10.10/443 (10.10.1.12/3306) (kjones) action = allowed host = show-splk10-demo-i-01dfea53c5bc7433d source = /var/log/firewall/cisco_asa.log6 sourcetype = cisco:asa
>	19/06/2026 16:09:24.000	Jun 19 16:09:24 fw-01 : %ASA-6-302013: Built outbound TCP connection 1605179675 for outside:141.98.10.20/29348 (185.246.128.161/43104) to inside:10.50.10.10/80 (10.50.10.110/1433) (pwalker) action = allowed host = show-splk10-demo-i-01dfea53c5bc7433d source = /var/log/firewall/cisco_asa.log6 sourcetype = cisco:asa
>	19/06/2026 16:09:24.000	Jun 19 16:09:24 fw-01 : %ASA-4-106023: Deny udp src outside:178.62.61.42/65281 dst inside:10.20.30.42/3389 by access-group "outside_access_in" [0x0, 0x0] action = blocked host = show-splk10-demo-i-01dfea53c5bc7433d source = /var/log/firewall/cisco_asa.log6 sourcetype = cisco:asa
>	19/06/2026 16:09:24.000	Jun 19 16:09:24 fw-01 : %ASA-4-106023: Deny udp src outside:61.177.173.45/16826 dst inside:10.50.10.102/80 by access-group "outside_access_in" [0x0, 0x0] action = blocked host = show-splk10-demo-i-01dfea53c5bc7433d source = /var/log/firewall/cisco_asa.log6 sourcetype = cisco:asa
>	19/06/2026 16:09:24.000	Jun 19 16:09:24 fw-01 : %ASA-4-106023: Deny tcp src outside:172.245.115.102/62420 dst inside:10.50.10.101/8080 by access-group "outside_access_in" [0x0, 0x0] action = blocked host = show-splk10-demo-i-01dfea53c5bc7433d source = /var/log/firewall/cisco_asa.log6 sourcetype = cisco:asa
>	19/06/2026 16:09:23.000	Jun 19 16:09:23 fw-01 : %ASA-4-106023: Deny tcp src outside:185.220.101.34/21026 dst inside:10.10.2.12/445 by access-group "outside_access_in" [0x0, 0x0] action = blocked host = show-splk10-demo-i-01dfea53c5bc7433d source = /var/log/firewall/cisco_asa.log10 sourcetype = cisco:asa

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34

Create a Table of All IP Addresses That Appear in Both the Web Server and Blocked Firewall Datasets

13. Run the following search over the **Last 60 minutes**:

```
index=firewall sourcetype=cisco:asa
```

You should see firewall events appearing. You will also see that Splunk has automatically extracted fields like **src_ip**, **dest_ip** and **action**. The firewall events also have a **clientip** field — the Splunk Add-on for Cisco ASA aliases the firewall data's **src_ip** field to **clientip** so both our web and firewall log sources use a common taxonomy.

>	22/06/2026 09:49:04.000	131.178.233.243 - - [22/Jun/2026 09:49:04] "GET /product.screen?uid=e36baaa2-8de1-4612-9362-5ad2b64272ee&product_id=Z5G-2&JSESSIONID=SD4SL5FF18ADFF5 HTTP/1.1" 200 3579 "http://www.buttercupenterprises.com/product.screen?product_id=BW-3" "Mozilla/5.0 (iPhone; CPU iPhone OS 14_0 like Mac OS X) AppleWebKit/603.1.30 Version/14.0 Mobile/15A5378a Safari/602.1 (compatible; bingbot/2.0; http://www.bing.com/bingbot.htm)" 756	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/weblogs/noise_apache.log23	sourcetype = access_combined	access_combined Web logs
>	22/06/2026 09:49:04.000	130.253.37.97 - - [22/Jun/2026 09:49:04] "GET /product.screen?uid=41e33984-df95-4626-82d0-01e5c78cec14&product_id=DFS-2&JSESSIONID=SD8SL10FF8ADFF7 HTTP/1.1" 200 2962 "http://www.buttercupenterprises.com/product.screen?product_id=BW-3" "Mozilla/5.0 (Linux; Android 12.0.0; SM-S901B Build/S908EXXU28VJA) AppleWebKit/537.36 Chrome/114.0.5735.131 Mobile Safari/537.36" 837	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/weblogs/noise_apache.log23	sourcetype = access_combined	
>	22/06/2026 09:49:04.000	Jun 22 09:49:04 fw-01 : %ASA-4-106023: Deny tcp src outside:61.177.173.45/24356 dst inside:10.20.30.51/1433 by access-group "outside_access_in" [0x0, 0x0]	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/firewall/cisco_asa.log10	sourcetype = cisco:asa	cisco:asa Firewall logs
>	22/06/2026 09:49:04.000	Jun 22 09:49:04 fw-01 : %ASA-4-106023: Deny tcp src outside:130.253.37.97/45648 dst inside:10.50.10.100/3389 by access-group "outside_access_in" [0x0, 0x0]	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/firewall/cisco_asa.log10	sourcetype = cisco:asa	
>	22/06/2026 09:49:04.000	Jun 22 09:49:04 fw-01 : %ASA-4-106023: Deny tcp src outside:213.32.105.182/46250 dst inside:10.20.30.40/5900 by access-group "outside_access_in" [0x0, 0x0]	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/firewall/cisco_asa.log10	sourcetype = cisco:asa	

14. Bring both log web and firewall data sources together in a single search by running the following search:

```
(index=main sourcetype=access_combined) OR  
(index=firewall sourcetype=cisco:asa action=blocked)
```

You will now see web-server hits and firewall blocks (look for "Deny") together.

Jun 19 13:02:50 fw-01 : %ASA-4-106023: Deny tcp src outside:31.184.236.55/33556 dst inside:192.168.1.30/443 by access-group "outside_access_in" [0x0, 0x0]	action = blocked	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/firewall/cisco_asa.log6	sourcetype = cisco:asa
Jun 19 13:02:50 fw-01 : %ASA-4-106023: Deny tcp src outside:45.146.166.42/33192 dst inside:10.10.2.11/8443 by access-group "outside_access_in" [0x0, 0x0]	action = blocked	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/firewall/cisco_asa.log6	sourcetype = cisco:asa
Jun 19 13:02:50 fw-01 : %ASA-4-106023: Deny tcp src outside:198.50.182.151/8167 dst inside:10.20.30.41/443 by access-group "outside_access_in" [0x0, 0x0]	action = blocked	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/firewall/cisco_asa.log6	sourcetype = cisco:asa
141.146.8.66 - - [19/Jun/2026 13:02:49] "GET /cart.do?action=addtocart&product_id=PP-5&JSESSIONID=SD2SL5FF2ADFF1 HTTP/1.1" 400 2207 "http://www.buttercupenterprises.com/product.screen?product_id=BW-3" "Mozilla/5.0 (iPhone; CPU iPhone OS 15_0 like Mac OS X) AppleWebKit/605.1.15 Version/15.0 Mobile/19A346 Safari/602.1" 786	action = addtocart	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/weblogs/noise_apache.log1	sourcetype = access_combined
128.241.220.82 - - [19/Jun/2026 13:02:49] "GET /cart.do?action=view&product_id=BS-2&JSESSIONID=SD6SL5FF5ADFF3 HTTP/1.1" 200 743 "http://www.buttercupenterprises.com/product.screen?product_id=CM-1" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_13_6) AppleWebKit/537.36 Chrome/114.0.5735.106 Safari/537.36" 704	action = view	host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/weblogs/noise_apache.log1	sourcetype = access_combined
131.178.233.243 - - [19/Jun/2026 13:02:49] "POST /product.screen?uid=988d54cf-45f5-4b92-b51b-6352e6002ec0&product_id=CM-1&JSESSIONID=SD10SL5FF2ADFF4 HTTP/1.1" 200 1621 "http://www.buttercupenterprises.com/product.screen?product_id=MCF-3" "Mozilla/5.0 (Windows; WOW64; rv:109.0) Gecko/20100101 Firefox/115.0" 628		host = show-spkl0-demo-i-01dfea53c5bc7433d	source = /var/log/weblogs/noise_apache.log1	sourcetype = access_combined

Now let's create a table of the results.

15. Use the `stats` command to count how many different data sources each clientip appears in:

```
(index=main sourcetype=access_combined) OR  
(index=firewall sourcetype=cisco:asa action=blocked)  
| stats dc(sourcetype) AS sources BY clientip
```

 ← Add this line

clientip	sources
1.16.0.0	1
125.17.14.100	2
128.241.220.82	1
130.253.37.97	2
103.244.36.171	1
104.244.79.143	1

We're only interested in client IP addresses that appear in both (i.e. 2) so use `where` command to filter by only clientips with 2 sources:

```
(index=main sourcetype=access_combined) OR  
(index=firewall sourcetype=cisco:asa action=blocked)  
| stats dc(sourcetype) AS sources BY clientip  
| where sources=2
```

 ← Add this line

This should now result in a small table of the suspect IPs – the ones that hit both the website and the firewall.

clientip	sources
125.17.14.100	2
130.253.37.97	2
141.146.8.66	2
175.45.176.66	2
175.45.178.19	2
194.215.205.19	2

Create a Cluster Map to show the geographic location of the suspicious IP addresses

Now we have the list of IP addresses we can plot these on a map.

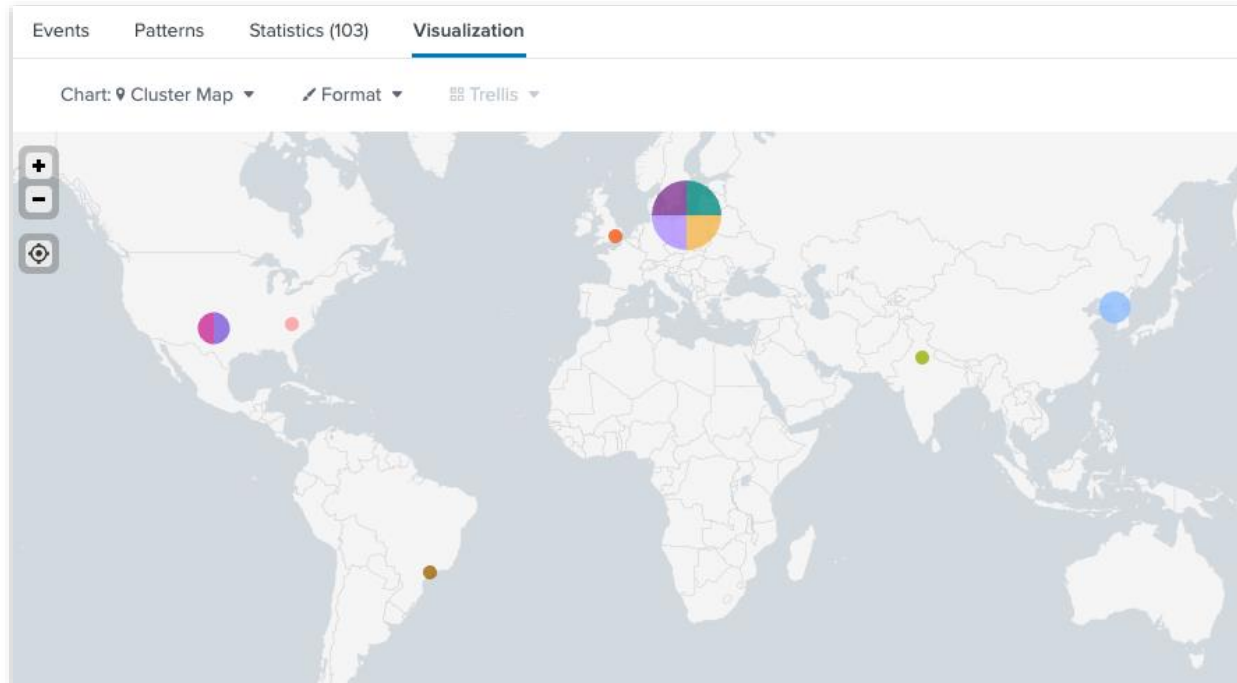
16. Use the `iplocation` and `geostats` commands to count the events by **City** (Note: 'City' is one of the fields in our data that's created when we use the `iplocation` command):

```
(index=main sourcetype=access_combined) OR  
(index=firewall sourcetype=cisco:asa action=blocked)  
| stats dc(sourcetype) AS sources BY clientip  
| where sources=2  
| iplocation clientip  
| geostats count BY City
```

 ← Add these lines

17. If it isn't selected already, click on the Visualization tab. For your visualization type, choose Cluster Map

You should now have a map showing the location of clients (IP addresses) that hit both the website and were blocked by the firewall.



18. Don't forget to add the resulting map to your dashboard and give your panel a name such as '**Security and Fraud – Potentially Malicious Traffic**'.

Challenge Tasks

The Cluster Map shows where the suspect IPs are coming from. The Security and Fraud team also wants to know what services the foreign attackers are most interested in, i.e. which destination ports are being hit hardest from outside the country.

Q1. Add a table next to the cluster map that shows the top 5 destination ports being attacked by source (client) IPs based outside the United States.

Hints:

- The first part of every Splunk search includes an implicit `search` command, so we don't need to use a `search` command at the start of our searches. However, in Splunk if we want to apply a search filter after a pipe (`" | "`) has been used – such as to filter out certain results – then we will need to specify the search command somewhere in our search query (i.e. `| search <search terms>`)
- **Note:** Remember that when searching, if we want to use a field to filter our results, we need to make sure the field exists at that point in our search – as we've seen today, some commands will add or remove fields as Splunk steps through our search query! Look at the commands you're using and remember which fields each command may be adding or removing from your data

✓ Challenge Task Solutions

The challenge task solutions are at the [end of this document](#).

Exercise 7 – Customize Your Dashboard

Description

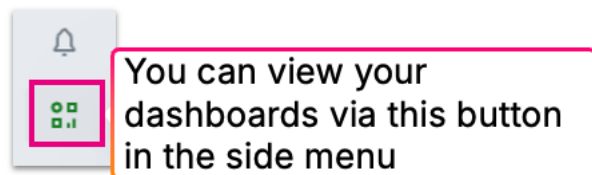
Having a dashboard with multiple panels is powerful, but the layout of your dashboard is also important to ensure that the information presented is clear and easy for users to consume.

The Buttercup Enterprises Marketing team has seen what we've built so far and have provided us with a custom background image that they would like us to use on our new dashboard. In this exercise we will upload the custom background image and rearrange our panels to work with the new background. Finally, we will configure each of our dashboard panels to use the global time picker so it's all ready to share with the business!

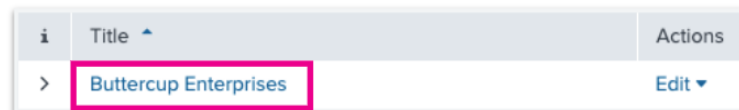
Steps

Add a Custom Background Image to Your Dashboard

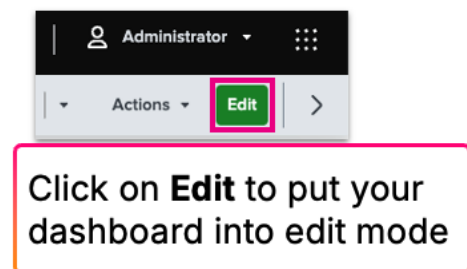
1. First, open your dashboard. To do that, click on the **Dashboards** icon in the side menu bar.



2. Click on the name of your dashboard to open it.



3. Click on the **Edit** button to put your dashboard into edit mode.



4. Locate the **Background** section and copy/paste the following image URL into the **Background image** box:

<https://splk.it/ButtercupBackground>

Paste the image URL here

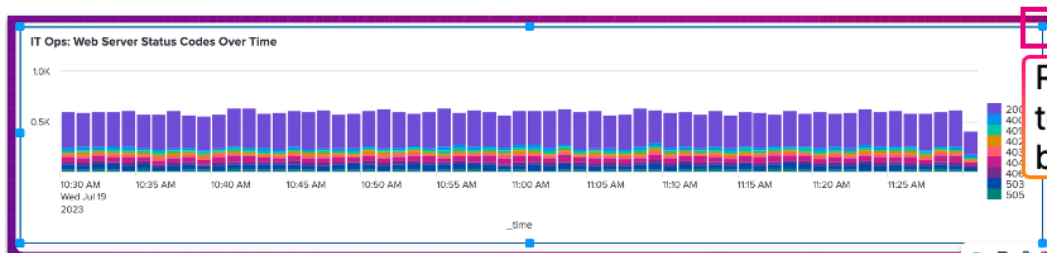
To upload the image, hit the Enter key on your keyboard.

To ensure that our custom image is contained within the dimensions of our dashboard, click on the dropdown list beneath the image preview and select '**Contain**'.

You can view a preview of the custom image here

Select **Contain**

5. Now click on each dashboard panel and drag the blue squares that appear around the edges of the panels to resize them to fit within the areas on your custom background image.

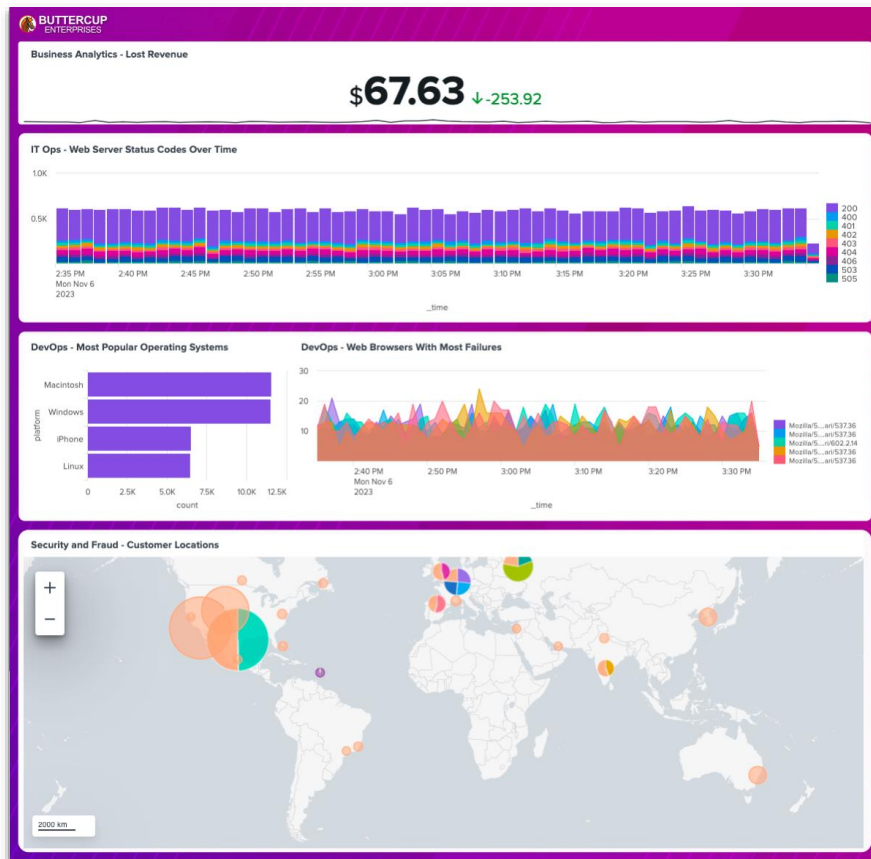


Resize your panels to fit the custom background

Be sure to click on **Save** when you've finished rearranging everything!



When you're finished, your dashboard should now look something like this:



6. Finally, since we have a nice colored background to show off we can set each panel to be transparent to help the color to shine through! To do this, click on a dashboard panel and in the Configuration panel on the right find the **Color and style** section.

Find the '**Background color**' option for your panel and change the color to be **transparent**. Repeat this step for each dashboard panel.

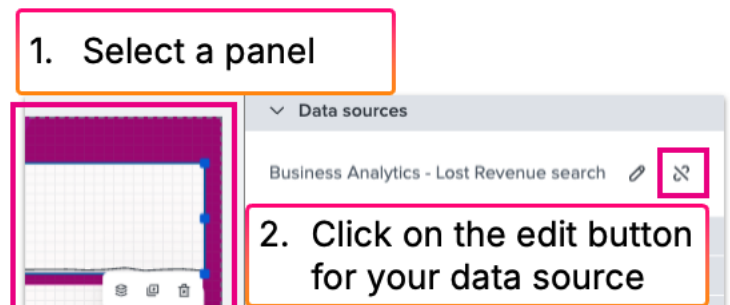
Note: The Cluster Map visualization has no background color option so you can ignore this panel.



Link Your Dashboard Panels to the Global Time Picker

The global time picker is included in all new dashboards by default and allows you to control the time range of all dashboard panels from a single place. Since each of our panels was using a static time range (e.g. **Last 60 minutes**) when we added them to our dashboard, we just need to switch each panel to use the global time picker instead.

7. With your dashboard in edit mode, click a dashboard panel and in the Configuration panel on the right find the **Data sources** section. Click on the pencil icon to edit the data source.



8. Each data source consists of an SPL query (i.e. a search) and a time range.

You will notice that your data source has a generic default name, but it is good practice to rename this, so the name is more unique and meaningful - this helps if you decide to reuse this data source for other dashboard panels in future.

Give the data source a new name. For simplicity, use the name of the dashboard panel. For example, if you're working with our Single Value visualization panel, use 'Lost Revenue' as the data source name.

For the time range select '**Input**'. This will then show 'Global Time Range (global_time)' in the dropdown list below - this means the data source will now use our dashboard's built-in ("global") time picker.

Click on **Apply and close** to save your panel changes

The screenshot shows the 'Edit data source' dialog box. The 'Data source name' field contains 'Business Analytics - Lost Revenue search'. The 'SPL query' field contains a query: `index=main sourcetype=access_combined action=purchase status>=400 | lookup product_codes.csv product_id | timechart sum(produ`. The 'Time range' dropdown is set to 'Input'. Below it, 'Global Time Range (global_time)' is shown. The 'Search refresh' is set to 'Off' and the 'Refresh interval' is set to '5'. Two callouts are present: '1. Give the panel's data source a name' pointing to the 'Data source name' field, and '2. Change the time range to Input' pointing to the 'Time range' dropdown.

Repeat steps 7 and 8 for all dashboard panels.

9. Now that you've linked all your panels to the global time picker, click on **Save** and then click on **View** to view your updated dashboard. Try changing the search time range for your dashboard by choosing different time ranges from the dropdown list.

All your panels should update to reflect the time setting.

The screenshot shows the 'Buttercup Enterprises' dashboard. At the top, there is a 'Global Time Range' dropdown menu set to 'Last 60 minutes'. Below this is a 'Presets' section with a grid of time range options. A callout box says 'Test your time picker by trying different time ranges!'. The grid is organized into four columns: 'Real-time', 'Relative', 'Other', and 'All time'. The 'Real-time' column includes '30 second window', '1 minute window', '5 minute window', '30 minute window', '1 hour window', and 'All time (real-time)'. The 'Relative' column includes 'Business week to date', 'Today', 'Week to date', 'Month to date', 'Year to date', 'Yesterday', 'Previous week', 'Previous business week', 'Previous month', and 'Previous year'. The 'Other' column includes 'Last 15 minutes', 'Last 60 minutes', 'Last 4 hours', 'Last 24 hours', 'Last 7 days', and 'Last 30 days'. The 'All time' column includes 'All time'.

Challenge Task Solutions

Below are suggested solutions to the challenge tasks contained in this lab guide. Don't worry if you used a slightly different method – there are often multiple ways of reaching the same result!

Start Searching in Splunk

Q1. How can we find events with a status of **200** that are not purchase events?

Solution:

```
status=200 action!=purchase
```

i NOT vs !=

status=200 **NOT** action=purchase will also work for this exercise, but this is not a good way of performing this query due to the way that the **NOT** operator works.

If you're interested to learn why, please see [Difference between != and NOT](#) in the Splunk docs for a full explanation of the differences between these two methods.

Q2. How can we find events where someone had an error when trying to either add an item or remove an item from their cart?

Solution:

```
index=main sourcetype=access_combined status>=400 (action=addtocart OR action=remove)
```

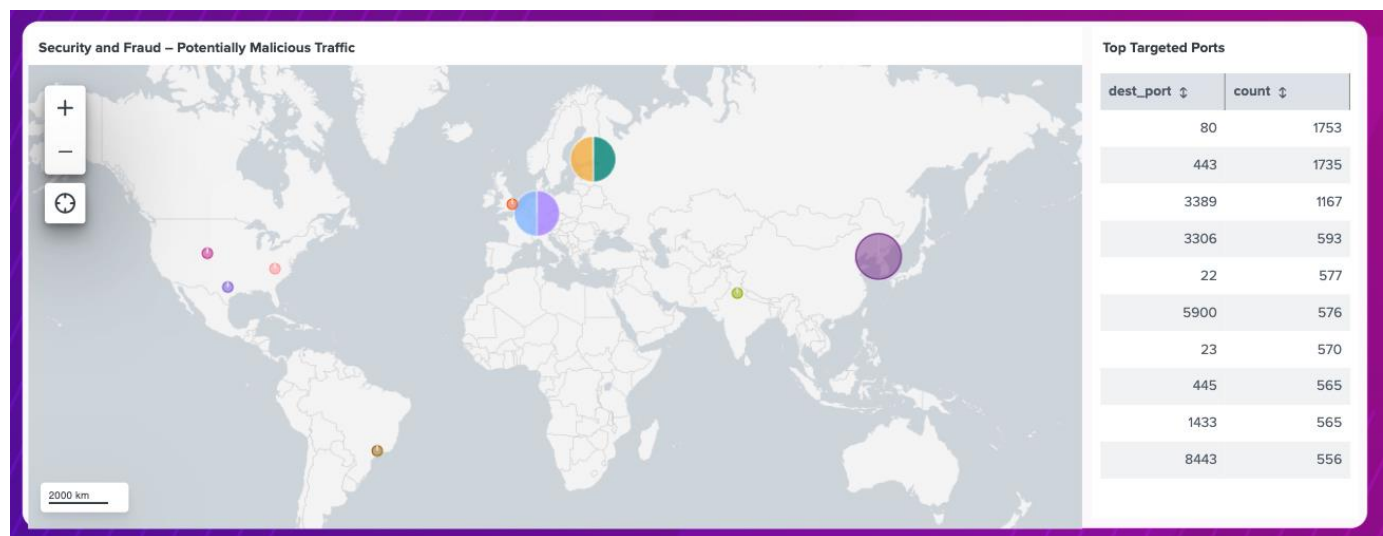
Exercise 6 – Security and Fraud team: Find suspicious actors hitting both the firewall and the website

Q1. Add a table next to the cluster map that shows the top 5 destination ports being attacked by source (client) IPs based outside the United States.

Solution:

```
index=firewall sourcetype=cisco:asa action=blocked
| iplocation clientip
| search Country!="United States"
| top dest_port limit=5 showperc=f
```

dest_port	count	percent
443	1758	19.106619
80	1754	19.063145
3389	1146	12.455168
8080	593	6.444952
5900	575	6.249321



Explanation:

1. Search for all blocked firewall traffic:

```
index=firewall sourcetype=cisco:asa action=blocked
```

2. Add geographic location data based on the **clientip** field:

```
index=firewall sourcetype=cisco:asa action=blocked
| iplocation clientip
```

← Add this line

3. Use the **search** command to filter by only events where the **Country** field is not equal to “United States”:

```
index=firewall sourcetype=cisco:asa action=blocked
| iplocation clientip
| search Country!="United States" ← Add this line
```

4. Use the [top](#) command to list the top 5 **dest_port** values seen in our data:

```
index=firewall sourcetype=cisco:asa action=blocked
| iplocation clientip
| search Country!="United States"
| top dest_port limit=5 ← Add this line
```

Play around with increasing the limit value if you want to see more ports.

5. Clean up the table by removing the percent column:

```
index=firewall sourcetype=cisco:asa action=blocked
| iplocation clientip
| search Country!="United States"
| top dest_port limit=5 showperc=f ← Add this parameter
```

dest_port	count
443	1758
80	1754
3389	1146
8080	593
5900	575

Add it to your dashboard!